

V3 empirical study · Instruction

Five models · four underlyings · four volatility regimes · 1.5-year fixed calibration · return-based calibration

What this report is

- Companion to the seven-model 10,000-path fixed-calibration study. This report keeps GBM, Modified GBM, GARCH, Merton, and GARCH–Merton, which are calibrated from lookback returns. Rankings are within this group; they are not compared here with option-implied Heston models.
- Companies are SPY, AAPL, MSFT, and AMZN. No new calibration or LSM. Every number is the stored cell from V3_1p5y_fixed_empirical_study.
- Option-implied companion: V3_1p5y_fixed_empirical_study_option_implied.pdf.
- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of 5 American-call pricing models tracks listed market prices most closely, and does the ranking change across companies and volatility regimes?
- In every table the BEST row is the lowest percentage RMSE. That row is bold and shaded green, and the Mark column says BEST.

Experimental design (held fixed for every model)

- Models: GBM, Modified GBM (Markov direction), GARCH(1,1), Merton jump-diffusion, GARCH–Merton.
- Modified GBM (Markov direction + split-normal magnitudes) is specified in V3_modified_gbm_model.pdf in this folder.
- Companies: SPY (primary), AAPL, MSFT, AMZN. Each name is calibrated and priced separately.
- Regimes (V3 one-year evaluation windows): Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- The late-cycle and COVID files share September 2019 (V3 notebook dates). All other regime days are disjoint. Contracts are built independently in each window.
- Calibration window: 1.5-year ending at the first session of each 1-year evaluation window. Rolling: none (one estimate, held for the whole window). Parameters estimated at t_0 are used only after t_0 .
- If history is shorter than the requested lookback, the estimator uses all available observations up to the update date (equity starts 2003-12-01; listed-option panels start 2008-01).

V3 empirical study · Instruction (continued)

Five models · four underlyings · four volatility regimes · 1.5-year fixed calibration · return-based calibration

Shared market sample and filters (not model-specific)

- For each company $\times$ regime, one listed-call sample is drawn once and reused by all five models in this report: nearest-ATM call on each Monday (next session if Monday is closed), DTE 7–60, listed expiry, as-of that date.
- Estimation filters, applied in order to every quote used for Heston/Heston–Merton calibration and to the LSM comparison set: (1) calls only, finite S , K , C ; (2) no-arbitrage $C \geq \max(0, S-K)$; (3) $7 \leq \text{DTE} \leq 60$; (4) $|S/K - 1| \leq 10\%$; (5) premium ≥ 0.05 , valid bid-ask with relative spread $\leq 50\%$ when those columns exist, volume ≥ 1 when present.
- Return-based models (GBM, Merton, GARCH, GARCH–Merton) never see option quotes in §4; they still price the same filtered LSM contracts in evaluation.

Model-correctness adjustments (V3)

- Heston / Heston–Merton: $(\kappa, \theta, \xi, \rho, v_0)$ are option-implied Fourier NLS (Albrecher little-trap), not Method A realized-variance moments. μ is the P-measure lookback mean of stock returns; LSM replaces $\mu \rightarrow r$.
- Euler ordering: the stock increment over $[t, t+\Delta t]$ uses the current variance v_t ; $v_{t+\Delta t}$ is updated afterwards. No look-ahead in the variance.
- LSM: Longstaff–Schwartz American call on the full path cloud ($n_{\text{paths}} = 10000$, seed 42, $n_{\text{steps}} = \text{DTE}$, $\Delta t = 1/252$). Paths are not averaged before stopping.

Metrics

- Primary: $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{\text{model}} - C_{\text{mkt}}) / C_{\text{mkt}})^2)}$. Ranking uses this number only.
- Secondary: $\text{MAE} = \text{mean} |C_{\text{model}} - C_{\text{mkt}}|$ (dollars). $\text{Bias \$} = \text{mean}(C_{\text{model}} - C_{\text{mkt}})$ (dollars). $\text{Bias\%} = 100 \times \text{mean}((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})$. Positive bias = model expensive vs market. Early-exercise fraction = mean share of LSM paths that stop before expiry.

Page 3 · Filters, sample, and no-look-ahead

Shared evaluation sample · n listed calls (identical for all five models in this report)

Regime	Window	SPY n	AAPL n	MSFT n	AMZN n
2008-2009 Crisis	2008-08-01 → 2009-07-31	50	50	50	50
2013-2014 Normal	2014-01-01 → 2014-12-31	50	50	50	50
2018-2019 Late-cycle	2018-10-01 → 2019-09-30	50	51	50	51
2019-2020 COVID	2019-09-01 → 2020-08-31	50	50	50	50

Filter funnel · last-step n in the 1.5-year lookback ending at each regime (processed call panel)

Ticker	Regime	Input*	No-arb	DTE 7-60	S/K-1 ≤10%	Liquidity
SPY	2008-2009	3359	3311	3311	3203	3203
SPY	2013-2014	12815	12548	12548	12405	12405
SPY	2018-2019	35222	35104	35104	34809	34809
SPY	2019-2020	42100	41961	41961	41379	41379
AAPL	2008-2009	685	685	685	652	652
AAPL	2013-2014	8925	8875	8875	8633	8633
AAPL	2018-2019	6309	6297	6297	6084	6084
AAPL	2019-2020	8010	7992	7992	7713	7713
MSFT	2008-2009	566	566	566	538	538
MSFT	2013-2014	3299	3237	3237	3175	3175
MSFT	2018-2019	7814	7752	7752	7583	7583
MSFT	2019-2020	6892	6852	6852	6655	6655
AMZN	2008-2009	465	465	465	442	442
AMZN	2013-2014	5715	5692	5692	5598	5598
AMZN	2018-2019	28809	28808	28808	28346	28346
AMZN	2019-2020	26766	26760	26760	26323	26323

*Input is the processed call panel restricted to (regime end – 1.5-year, regime end]. Processed files already drop many raw quotes; the funnel above is the additional V3 research filter.

No look-ahead. Monthly parameters at update date t use only equity returns and listed quotes with trading_date $\leq t$. LSM prices the Monday contract using the latest calibration row with date $\leq$ that Monday. Quotes are as-of the grid timestamp (same session, else the prior session).

SPY 2008–2009 uses bid-ask mid when the source mark is \$0.01 (a GitHub field error, not missing trades). After that repair the processed panel has weekly coverage, so n matches AAPL/MSFT (~50). All five models in this report still share that exact sample.

Heston / Heston-Merton calibration further subsamples the filtered lookback surface to ≤ 24 contracts (maturity $\times$ moneyness strata). GBM / Merton / GARCH / GARCH-Merton estimate from lookback log returns only (jumps: 3σ residual threshold where applicable).

Page 4 · Table block · SPY · five models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table SPY-1 SPY · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	39.85	1.5118	-1.3570	-29.27	25.5%	3
Modified GBM	43.13	1.6255	-1.5635	-35.30	26.1%	5
GARCH	35.72	1.3442	-0.8523	-14.47	21.6%	2
Merton	39.86	1.5142	-1.3598	-29.38	24.3%	4
GARCH-Merton	35.69	1.3398	-0.8186	-13.54	20.8%	1

Table SPY-2 SPY · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	40.10	0.7145	+0.5358	+27.51	26.2%	2
Modified GBM	41.67	0.7464	+0.5698	+29.03	26.1%	4
GARCH	40.43	0.7184	+0.5469	+27.93	23.4%	3
Merton	39.30	0.7016	+0.5143	+26.61	24.2%	1
GARCH-Merton	45.88	0.8274	+0.6986	+34.18	22.4%	5

Table SPY-3 SPY · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	23.62	0.8840	-0.1277	+1.81	20.8%	4
Modified GBM	23.03	0.8902	-0.2283	-0.59	22.2%	2
GARCH	25.80	1.0497	-0.8867	-16.54	17.2%	5
Merton	23.41	0.8884	-0.1839	+0.37	14.3%	3
GARCH-Merton	22.86	0.9032	-0.4350	-5.93	16.7%	1

Table SPY-4 SPY · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	57.52	2.9208	-0.5111	+18.64	24.3%	2
Modified GBM	59.99	2.9847	-0.3345	+21.88	23.9%	3
GARCH	63.77	3.0315	-0.0942	+26.41	21.2%	4
Merton	56.08	2.9267	-0.6252	+16.46	20.8%	1
GARCH-Merton	70.25	3.1595	+0.2869	+33.51	20.3%	5

Page 5 · Table block · Company summary · five models × four regimes

Each number is the arithmetic mean of AAPL, MSFT, AMZN. Same columns as the SPY page. Bold green row = lowest mean RMSE%. 1.5-year lookback, rolling=none, n_paths=100

Table Co-1 Company summary · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · mean of AAPL, MSFT, AMZN · $\bar{n} = 50$

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	32.08	1.0927	-0.1692	-4.59	23.9%	2
Modified GBM	30.51	1.0614	-0.4437	-9.68	25.3%	1
GARCH	36.12	1.1892	+0.3125	+11.82	17.6%	4
Merton	32.29	1.0895	-0.2081	-5.80	20.8%	3
GARCH-Merton	43.19	1.3542	+0.6781	+21.13	17.4%	5

Table Co-2 Company summary · 2013-2014 Normal
2014-01-01 → 2014-12-31 · mean of AAPL, MSFT, AMZN · $\bar{n} = 50$

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	47.76	2.2220	+1.4315	+36.60	24.6%	3
Modified GBM	44.74	2.0480	+1.2472	+33.61	25.7%	2
GARCH	48.16	1.9647	+1.3374	+37.70	20.5%	4
Merton	43.28	2.0864	+1.2262	+31.81	19.3%	1
GARCH-Merton	66.53	2.7630	+2.5061	+57.71	17.5%	5

Table Co-3 Company summary · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · mean of AAPL, MSFT, AMZN · $\bar{n} = 51$

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	20.48	4.6263	-2.0225	-2.87	23.3%	3
Modified GBM	20.18	4.6410	-1.3699	-3.86	23.8%	1
GARCH	20.21	4.6606	-1.1397	-1.61	19.0%	2
Merton	20.66	4.7540	-2.9521	-5.61	17.9%	4
GARCH-Merton	23.79	5.1311	+1.3847	+8.36	16.6%	5

Table Co-4 Company summary · 2019-2020 COVID
2019-09-01 → 2020-08-31 · mean of AAPL, MSFT, AMZN · $\bar{n} = 50$

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	48.08	10.4481	+4.7656	+23.77	24.2%	3
Modified GBM	48.43	10.8481	+5.6520	+23.33	24.2%	4
GARCH	47.17	9.5757	+0.8353	+22.31	20.5%	2
Merton	46.54	10.3044	+4.2731	+21.84	21.0%	1
GARCH-Merton	55.47	10.1345	+2.6265	+32.21	19.4%	5

RMSE%, MAE, Bias \$, Bias%, and early-exercise fraction are each averaged across AAPL, MSFT, AMZN. This is not a pooled re-pricing of a combined contract sample. Ranking uses the mean RMS

Page 6 · Table block · AAPL · five models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table AAPL-1 AAPL · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	34.70	1.7325	-0.2012	+2.49	23.4%	3
Modified GBM	32.72	1.6706	-0.6554	-4.84	24.4%	1
GARCH	40.04	1.9147	+0.4172	+12.21	17.2%	4
Merton	34.26	1.7225	-0.2873	+0.95	21.0%	2
GARCH-Merton	43.53	2.0289	+0.7015	+16.92	16.7%	5

Table AAPL-2 AAPL · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	57.70	3.8194	+3.8194	+51.38	24.4%	4
Modified GBM	51.10	3.3484	+3.3484	+44.85	25.7%	2
GARCH	45.96	2.9301	+2.9021	+39.06	20.4%	1
Merton	52.75	3.5099	+3.5099	+46.69	20.4%	3
GARCH-Merton	67.13	4.6054	+4.6054	+61.25	17.2%	5

Table AAPL-3 AAPL · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	16.77	0.8809	-0.5584	-7.11	23.8%	2
Modified GBM	16.88	0.8919	-0.5478	-6.91	24.5%	3
GARCH	16.17	0.8273	-0.3825	-4.32	19.9%	1
Merton	17.13	0.9184	-0.6295	-8.25	21.1%	4
GARCH-Merton	17.34	0.8370	+0.0909	+3.31	19.3%	5

Table AAPL-4 AAPL · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	41.39	3.1111	+0.6446	+18.83	24.0%	2
Modified GBM	43.50	3.2399	+0.9173	+21.54	24.2%	3
GARCH	54.11	3.9481	+2.2445	+34.15	20.1%	4
Merton	39.55	3.0542	+0.3844	+16.28	20.1%	1
GARCH-Merton	68.46	5.0306	+3.8961	+50.08	18.7%	5

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths e

Page 7 · Table block · MSFT · five models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table MSFT-1 MSFT · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	29.17	0.3120	-0.2729	-19.88	25.5%	2
Modified GBM	28.96	0.3119	-0.2681	-19.42	26.3%	1
GARCH	29.41	0.2879	+0.0218	+8.10	19.6%	3
Merton	30.20	0.3213	-0.2903	-21.66	23.7%	4
GARCH-Merton	32.88	0.2963	+0.0849	+14.04	20.0%	5

Table MSFT-2 MSFT · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	62.49	0.4284	+0.4262	+52.88	24.7%	3
Modified GBM	60.58	0.4138	+0.4100	+51.07	26.0%	2
GARCH	73.22	0.5232	+0.5232	+63.91	21.0%	4
Merton	55.05	0.3688	+0.3615	+45.40	17.6%	1
GARCH-Merton	96.85	0.7307	+0.7307	+87.77	17.1%	5

Table MSFT-3 MSFT · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	21.68	0.5626	+0.0129	+3.25	23.0%	4
Modified GBM	19.94	0.5524	-0.2155	-3.65	25.4%	1
GARCH	20.87	0.5485	-0.0844	+0.24	18.6%	2
Merton	21.14	0.5548	-0.0675	+0.85	17.2%	3
GARCH-Merton	25.36	0.6383	+0.2795	+11.22	16.3%	5

Table MSFT-4 MSFT · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	45.98	1.7242	+0.0264	+16.93	24.8%	4
Modified GBM	41.04	1.7043	-0.3996	+8.84	25.5%	1
GARCH	45.83	1.7300	+0.0064	+16.56	20.1%	3
Merton	45.04	1.7309	-0.0492	+15.51	21.8%	2
GARCH-Merton	53.39	1.8833	+0.4719	+25.84	18.4%	5

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths e

Page 8 · Table block · AMZN · five models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table AMZN-1 AMZN · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	32.38	1.2336	-0.0334	+3.64	22.9%	2
Modified GBM	29.85	1.2017	-0.4075	-4.77	25.2%	1
GARCH	38.93	1.3651	+0.4984	+15.15	15.9%	4
Merton	32.42	1.2247	-0.0467	+3.32	17.7%	3
GARCH-Merton	53.16	1.7374	+1.2478	+32.42	15.6%	5

Table AMZN-2 AMZN · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	23.09	2.4183	+0.0489	+5.54	24.8%	3
Modified GBM	22.52	2.3818	-0.0168	+4.90	25.5%	2
GARCH	25.31	2.4409	+0.5868	+10.13	19.9%	4
Merton	22.03	2.3804	-0.1930	+3.35	19.9%	1
GARCH-Merton	35.61	2.9530	+2.1822	+24.10	18.2%	5

Table AMZN-3 AMZN · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	22.97	12.4353	-5.5218	-4.77	22.9%	1
Modified GBM	23.72	12.4787	-3.3465	-1.02	21.5%	4
GARCH	23.59	12.6060	-2.9523	-0.74	18.5%	2
Merton	23.70	12.7887	-8.1593	-9.42	15.5%	3
GARCH-Merton	28.66	13.9181	+3.7835	+10.56	14.3%	5

Table AMZN-4 AMZN · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	56.86	26.5091	+13.6258	+35.54	23.7%	4
Modified GBM	60.74	27.6000	+16.4384	+39.62	22.9%	5
GARCH	41.58	23.0490	+0.2550	+16.22	21.4%	1
Merton	55.02	26.1281	+12.4842	+33.71	21.1%	3
GARCH-Merton	44.57	23.4896	+3.5116	+20.71	21.1%	2

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths e

Page 7 · Summary tables · RMSE% across companies and regimes

Table S1 · Overall ranking (mean RMSE% over 16 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 16 cells best
1	Merton	36.68	36.78	5
2	Modified GBM	37.46	36.88	5
3	GBM	37.89	37.28	1
4	GARCH	38.80	39.48	3
5	GARCH-Merton	46.35	44.05	2

Table S2 · Wins by company (lowest RMSE% in that company×regime)

Model	SPY wins	AAPL wins	MSFT wins	AMZN wins	Total
GBM	0	0	0	1	1
Modified GBM	0	1	3	1	5
GARCH	0	2	0	1	3
Merton	2	1	1	1	5
GARCH-Merton	2	0	0	0	2

Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT / AMZN)

Model	2008-2009 Crisis	2013-2014 Normal	2018-2019 Late-cycle	2019-2020 COVID	Mean of 4
GBM	34.03	45.85	21.26	50.44	37.89
Modified GBM	33.66	43.97	20.89	51.32	37.46
GARCH	36.02	46.23	21.61	51.32	38.80
Merton	34.18	42.28	21.35	48.92	36.68
GARCH-Merton	41.32	61.37	23.55	59.17	46.35

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).

Table S4.1 · SPY

Model	Mean RMSE%	Median	# regimes best
GBM	40.27	39.98	0
Modified GBM	41.95	42.40	0
GARCH	41.43	38.07	0
Merton	39.66	39.58	2
GARCH-Merton	43.67	40.78	2

Table S4.2 · AAPL

Model	Mean RMSE%	Median	# regimes best
GBM	37.64	38.04	0
Modified GBM	36.05	38.11	1
GARCH	39.07	43.00	2
Merton	35.92	36.90	1
GARCH-Merton	49.12	55.33	0

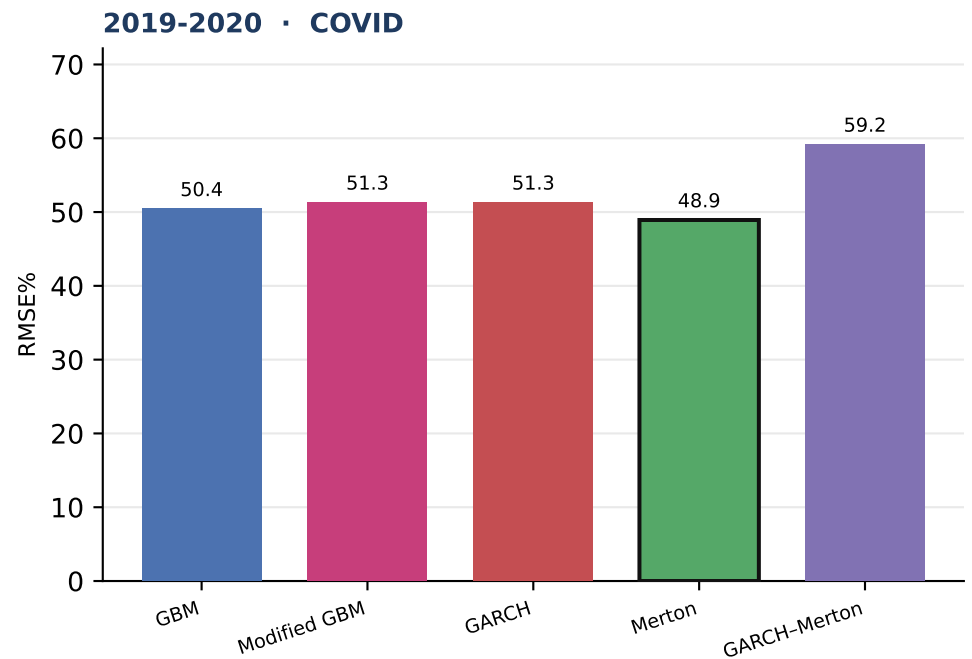
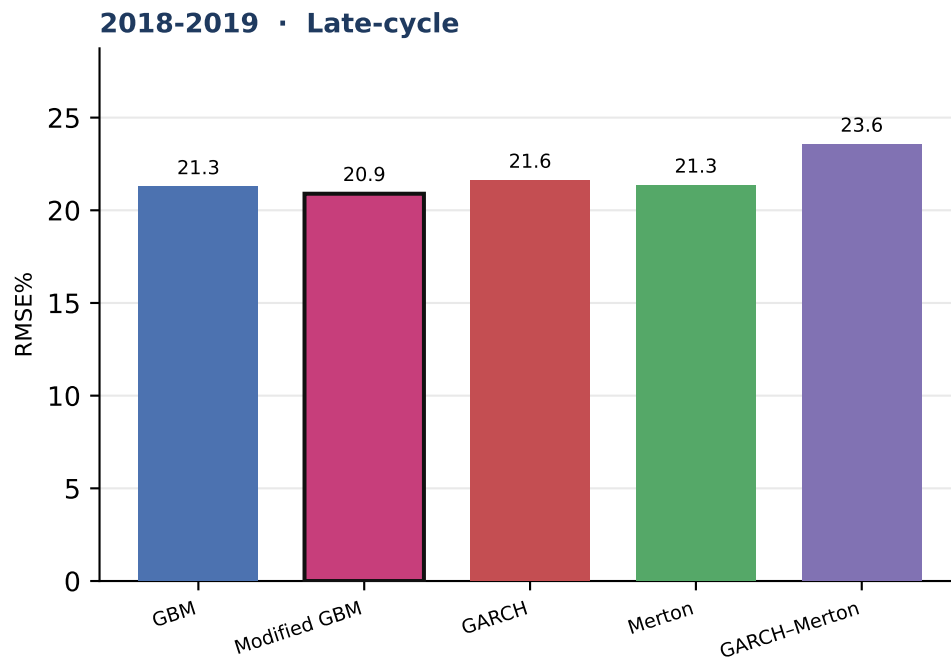
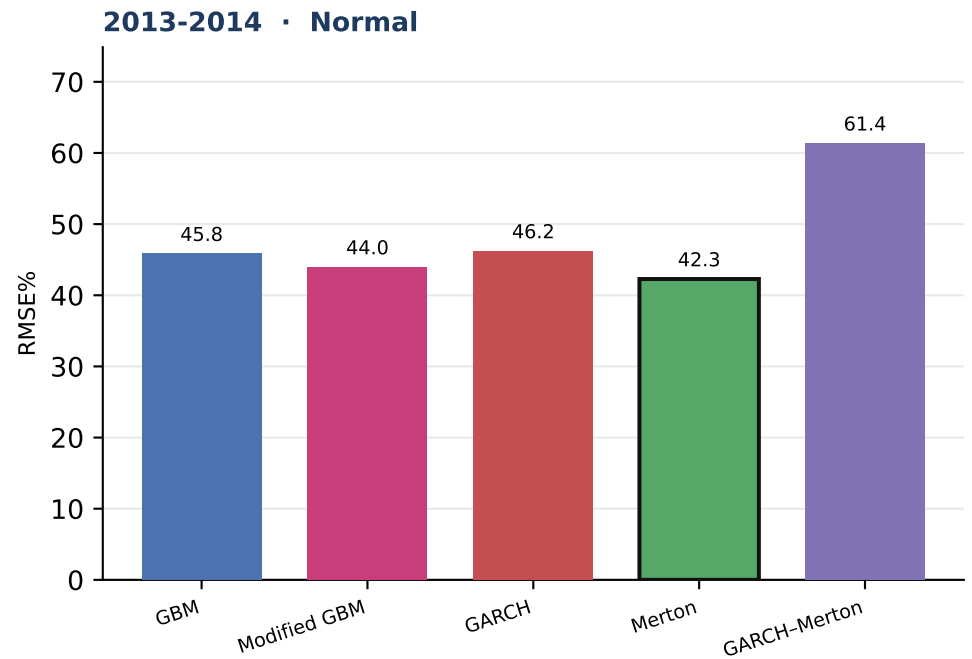
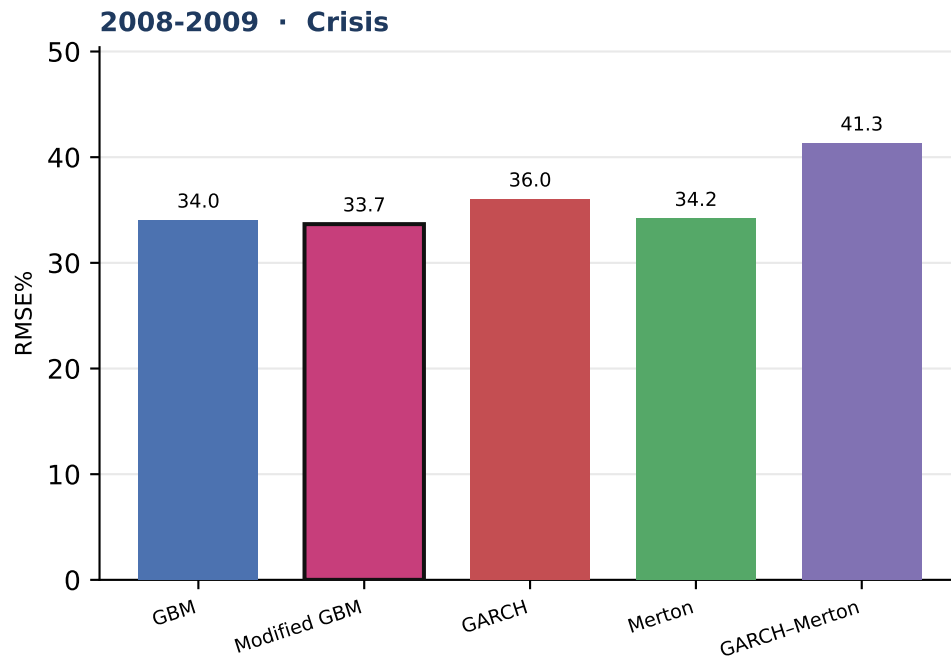
Table S4.3 · MSFT

Model	Mean RMSE%	Median	# regimes best
GBM	39.83	37.57	0
Modified GBM	37.63	35.00	3
GARCH	42.33	37.62	0
Merton	37.86	37.62	1
GARCH-Merton	52.12	43.14	0

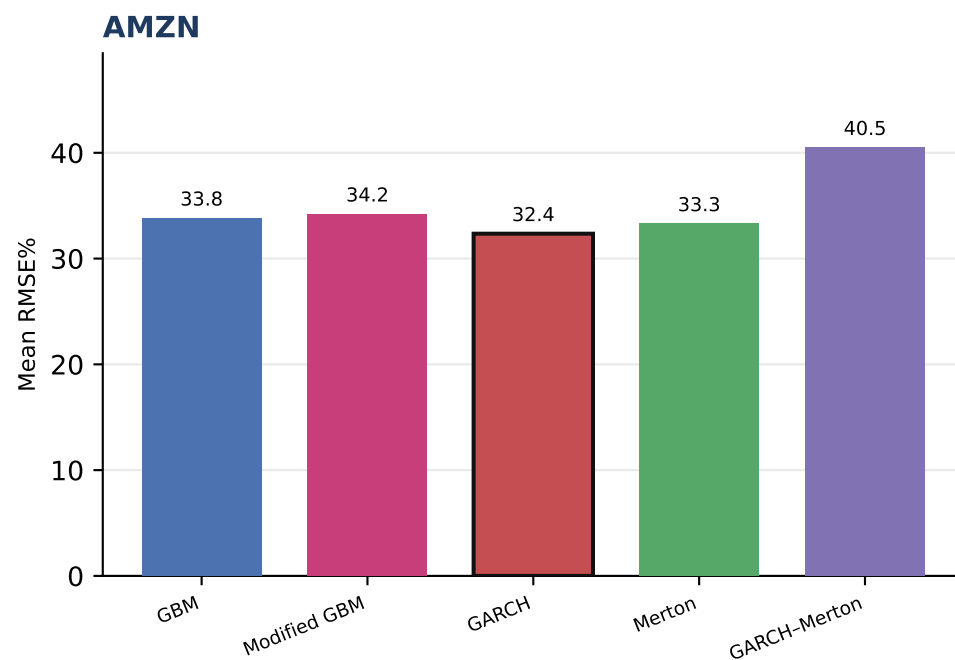
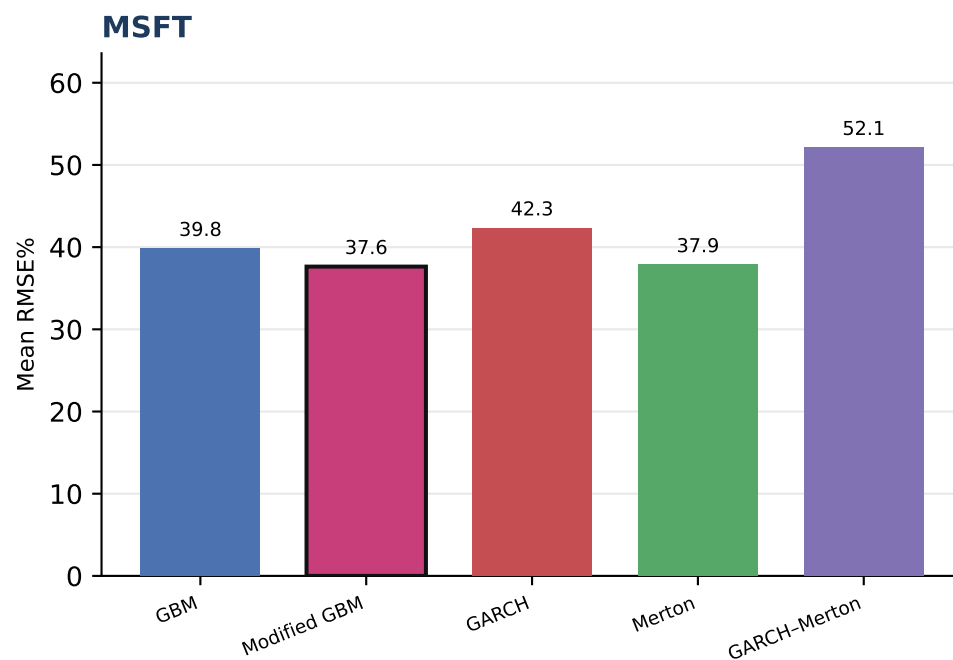
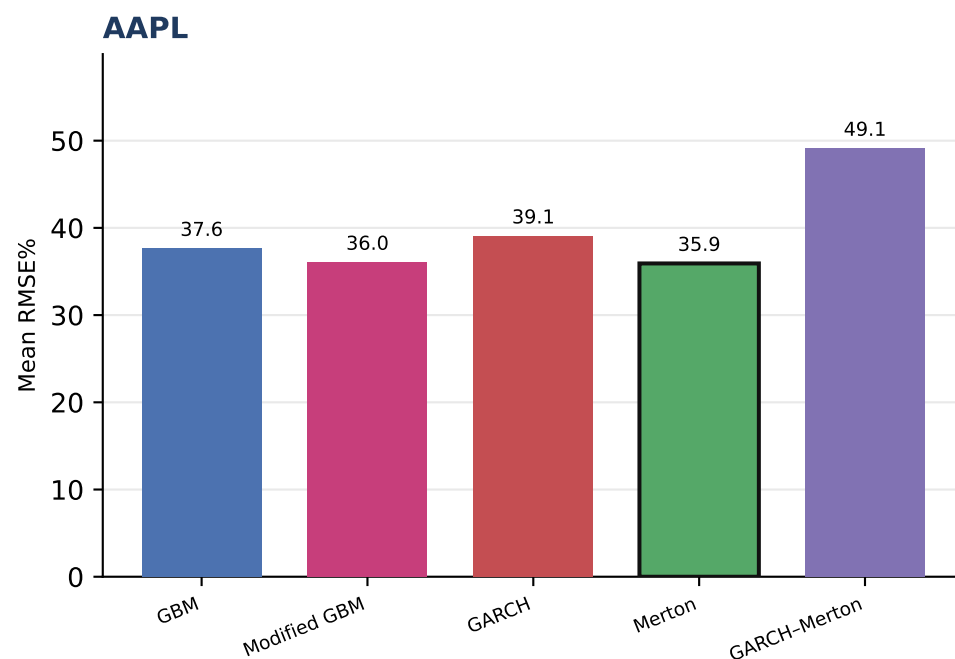
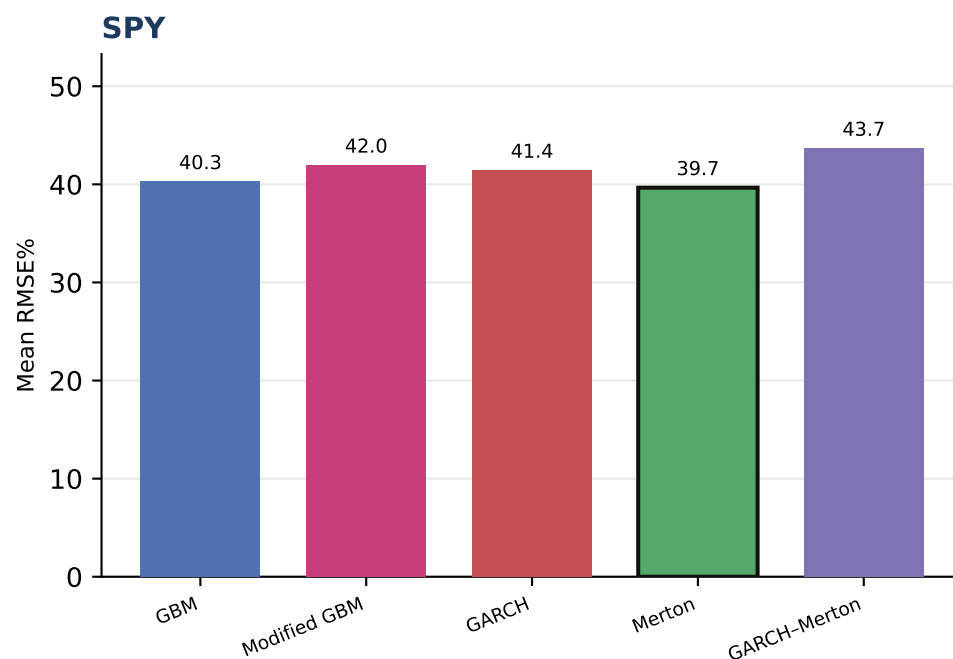
Table S4.4 · AMZN

Model	Mean RMSE%	Median	# regimes best
GBM	33.82	27.73	1
Modified GBM	34.21	26.79	1
GARCH	32.35	32.12	1
Merton	33.29	28.06	1
GARCH-Merton	40.50	40.09	0

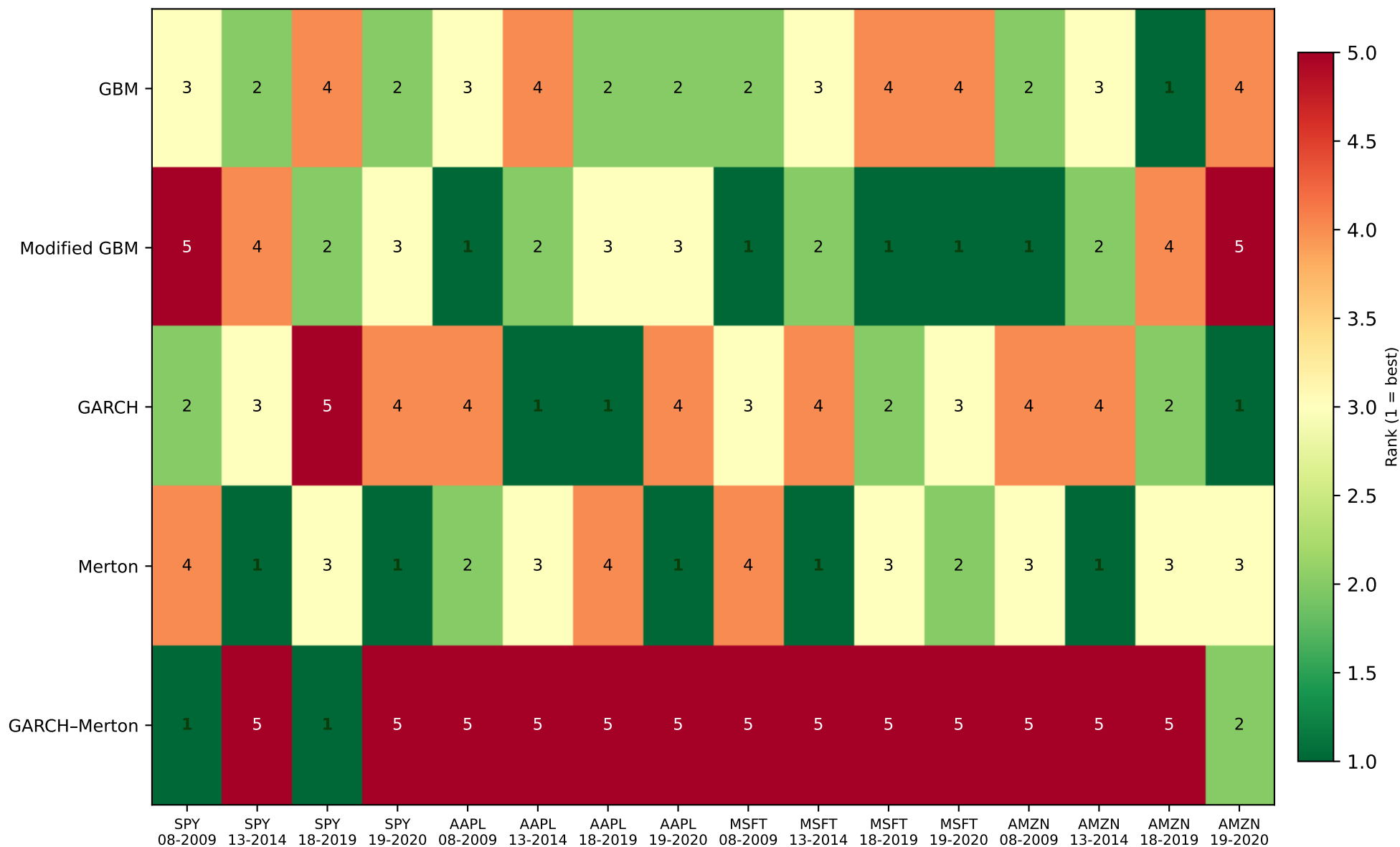
Each panel is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.



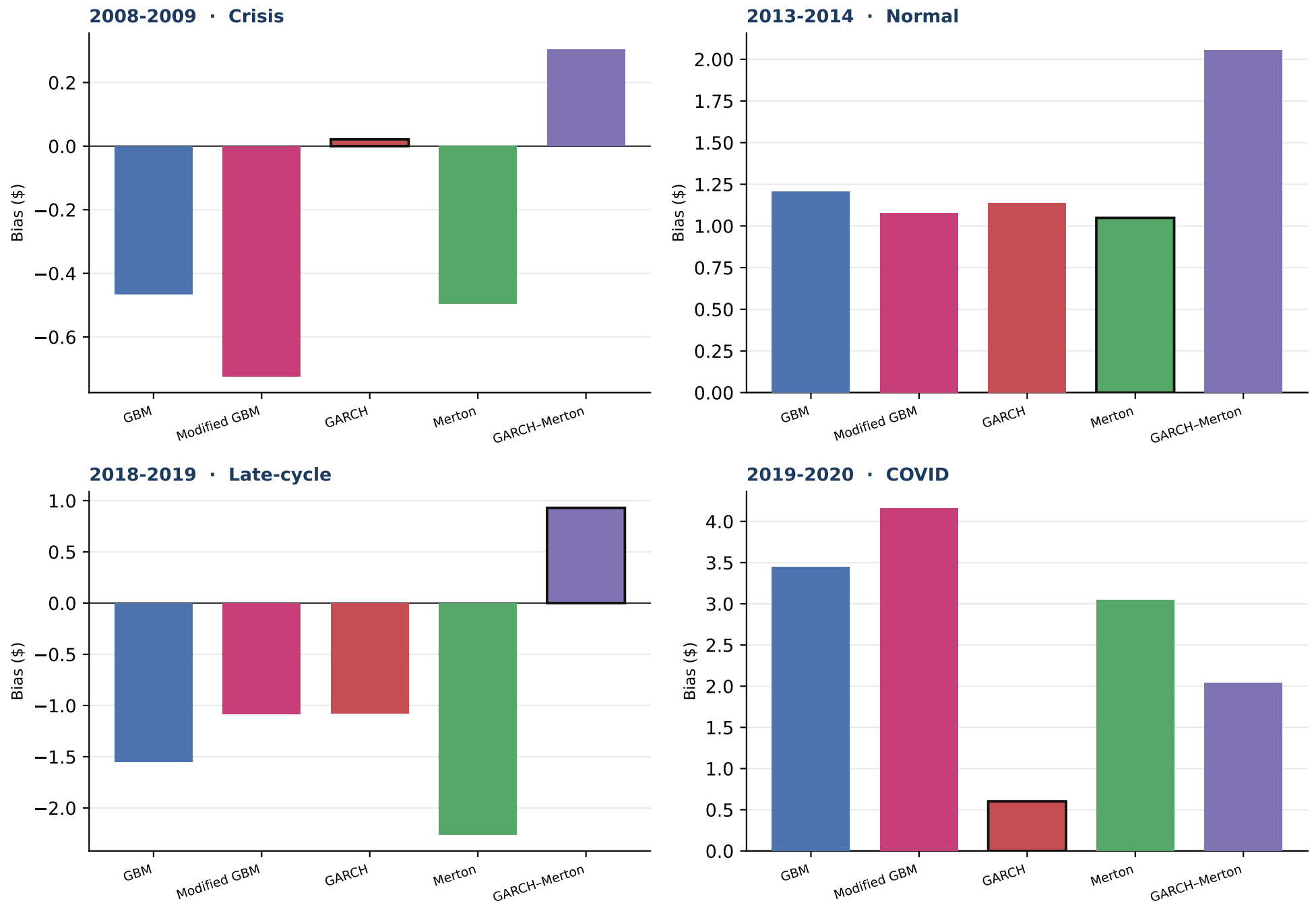
Outlined bar = lowest mean RMSE% in that regime. Equal-weight mean of the four companies.



Outlined bar = lowest mean RMSE% for that company.



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.



Outlined bar = closest to zero bias (not the RMSE ranking). Positive = model expensive vs listed quote.

Which models perform best, and does the ranking change?

- On mean percentage RMSE across the 16 company×regime cells, Merton is the best overall model (mean RMSE% = 36.68; 5 of 16 cells).
- Second overall is Modified GBM (mean RMSE% = 37.46; 5 cells).
- The ranking is not stable. Best-in-cell models across the 16 tables: GARCH, GARCH-Merton, GBM, Merton, Modified GBM.
- SPY: lowest mean RMSE% is Merton (39.66); wins 2 of 4 regimes.
- AAPL: lowest mean RMSE% is Merton (35.92); wins 1 of 4 regimes.
- MSFT: lowest mean RMSE% is Modified GBM (37.63); wins 3 of 4 regimes.
- AMZN: lowest mean RMSE% is GARCH (32.35); wins 1 of 4 regimes.
- 2008-2009 (Crisis): best mean RMSE% is Modified GBM (33.66).
- 2013-2014 (Normal): best mean RMSE% is Merton (42.28).
- 2018-2019 (Late-cycle): best mean RMSE% is Modified GBM (20.89).
- 2019-2020 (COVID): best mean RMSE% is Merton (48.92).
- No-jump models as a group have lower mean RMSE% (38.05) than the jump models in this report (41.52).
- For the overall winner (Merton), crisis RMSE% is 34.18 vs 42.28 in the 2014 normal year — absolute pricing error is regime-dependent even when the ranking is not.
- These conclusions are conditional on the V3 filters, 1.5-year fixed calibration, Monday ATM sample, LSM with 10,000 paths, and American-call quotes only. They are not a statement about European options or other tenors.

How to read the 12 detailed tables

- Each of Tables SPY-1...MSFT-4 uses exactly the same option contracts for all five models in this report. Differences are the dynamics, not the sample.
- BEST is always lowest RMSE%. MAE, Bias \$, Bias%, and early-exercise are reported, not used for the ranking.
- A model that is BEST on RMSE% can still be biased (systematically expensive or cheap).

Early-exercise fractions are a model implication, not an accuracy score. American calls on these tenors often have modest early-exercise value.

Notebook: [Reproduction of V3 Models](#) · Engine: [V3-Models_result/scripts/run_v3_1p5y_10k_fixed_empirical_study.py](#) · [payload.json](#) is the shared number store.